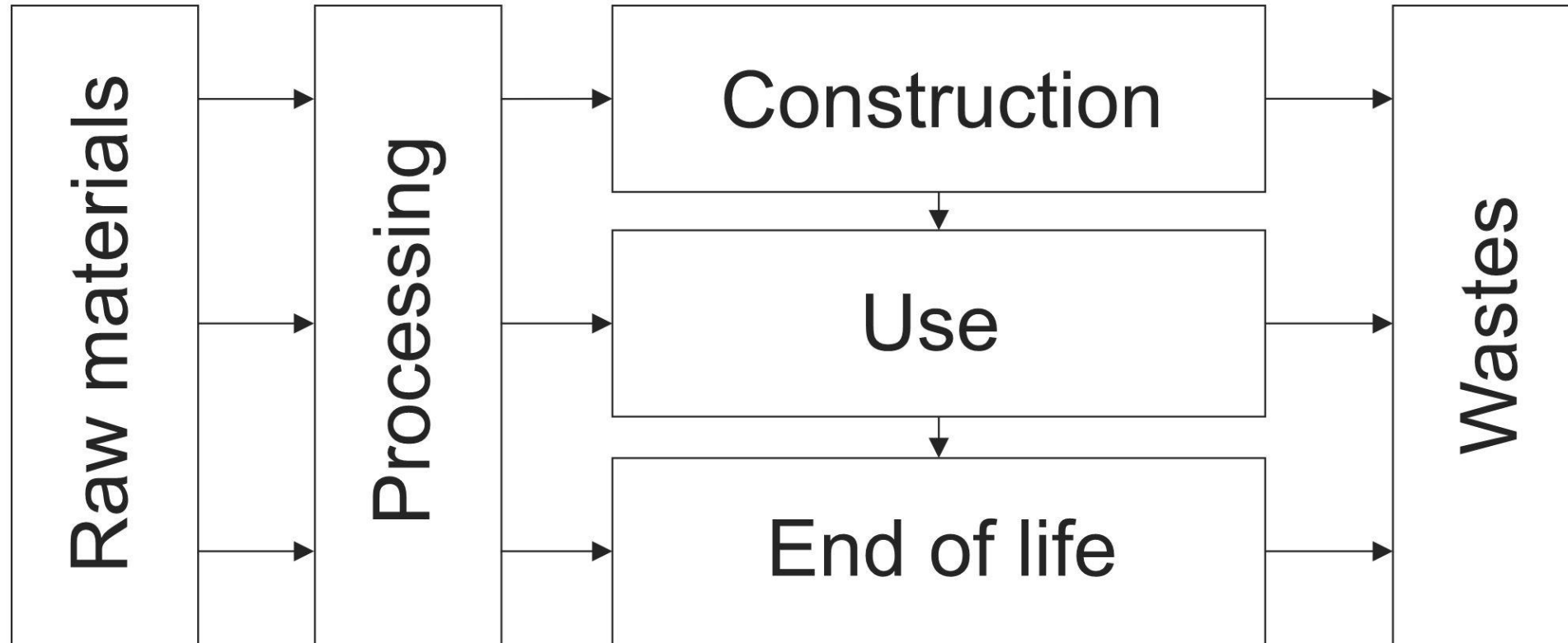


LCA 4 – Developing a life cycle flow chart

ENGEN370

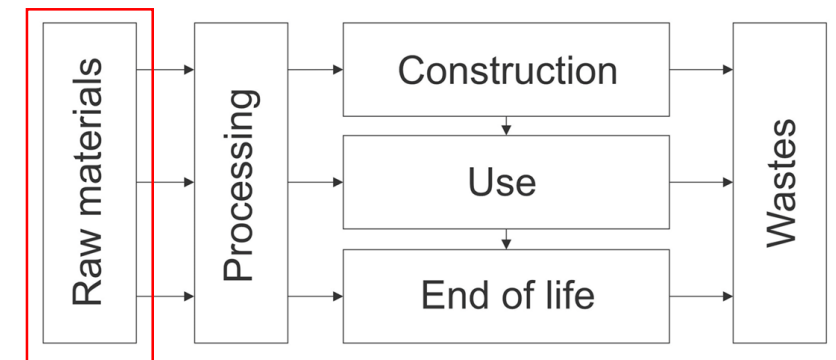
Mark Lay

Common features of LCA flow charts



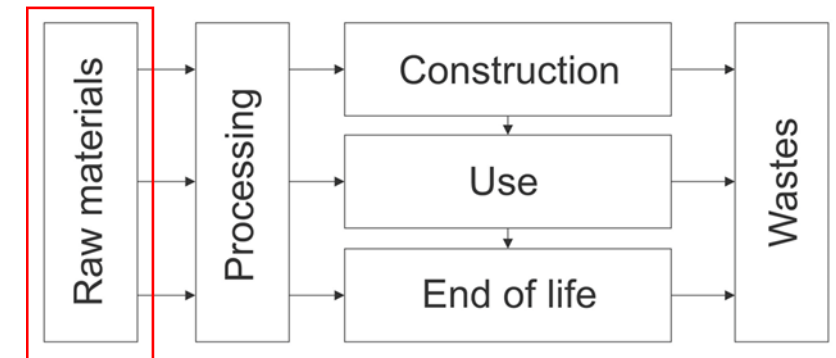
Raw materials

- Raw materials consist of anything that is used to make your product, process or infrastructure, and the extraction processes to recover those materials
- E.g. for concrete – aggregate, gravel, cement (limestone, sand, gypsum), plasticiser, water
- E.g. for a reinforced concrete wall – concrete, steel reinforcing (iron ore, coke), form-work (wood, nails, screws), spacers (plastic)
- E.g. for batteries – lithium salts, plastic, copper
- E.g. for PVC pipes – oil, salt



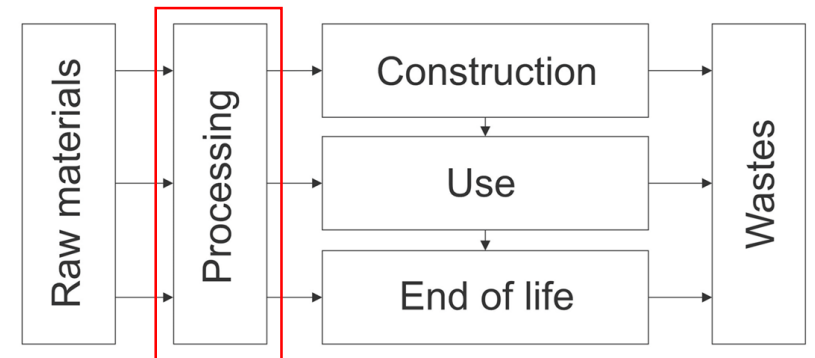
Raw material extraction

- E.g. aggregate, gravel, limestone, sand, gypsum – from a quarry (removing over-burden, digging up the material, crushing, sizing)
- E.g. oil – drilling, or digging if shale oil
- E.g. lithium salts from brine lakes – drilling, pumping, evaporation
- E.g. wood – growing and chopping down trees



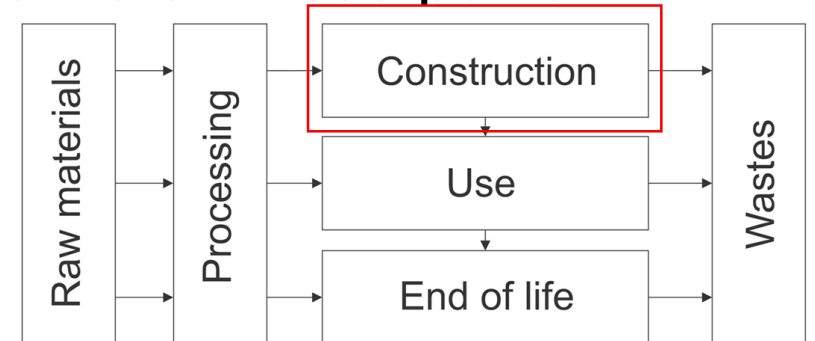
Processing

- Processing involves any step to process the raw materials into a useful form
- E.g. PVC – oil distillation, cracking, refining to produce ethylene, electrolysis of salt to produce chlorine, chlorination of ethylene to produce vinyl chloride, cross linking of vinyl chloride to produce PVC
- Cement – pyrolysis of lime to produce calcium oxide, mixing with gypsum and sand to produce cement



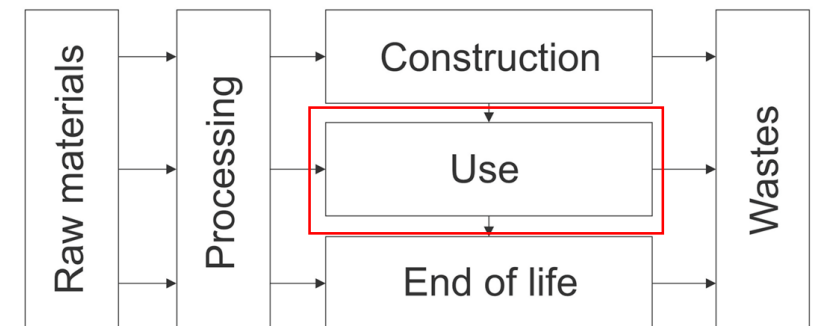
Construction

- This involves any step in the fabrication of your product, plant, or infrastructure
- E.g. a shopping mall: demolition, earth moving, piling, stormwater, water, sewage pipe installation, foundations, steel frame installation, roofing, flooring, cladding, windows, fire suppression, ventilation, heating, air conditioning, lifts, escalators, bathrooms, electric cabling, insulation, internal walls, lights, fittings, painting, carpets (in this case, you might like to include the manufacture of some of these parts under processing)



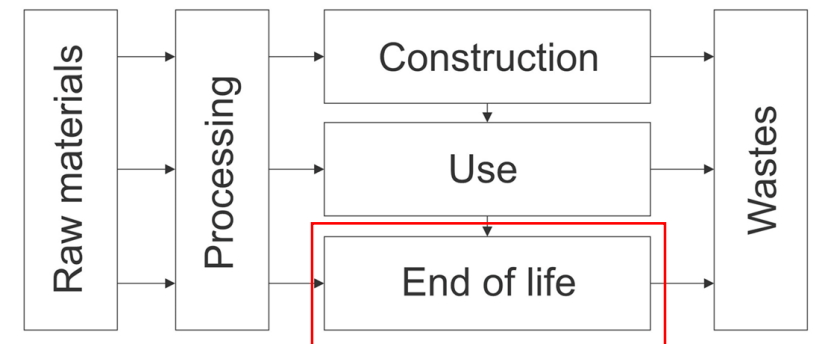
Use

- This involves day to day use of the product, plant or infrastructure, and maintenance
- E.g. car – fuel consumption, oil consumption, tire replacement, coolant use, windscreen wipers, batteries, catalytic converter, replacement of ball joints, push rods, shocks



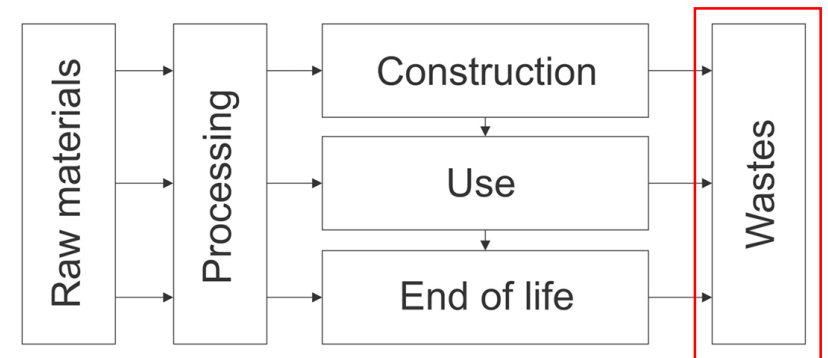
End of life

- This includes any process that occurs once your product, process or infrastructure has reached the end of its useful life
- E.g. shopping mall: demolition, recycling of materials, disposal of materials, rehabilitation
- E.g. car: stripping for parts, crushing, recycling, disposal



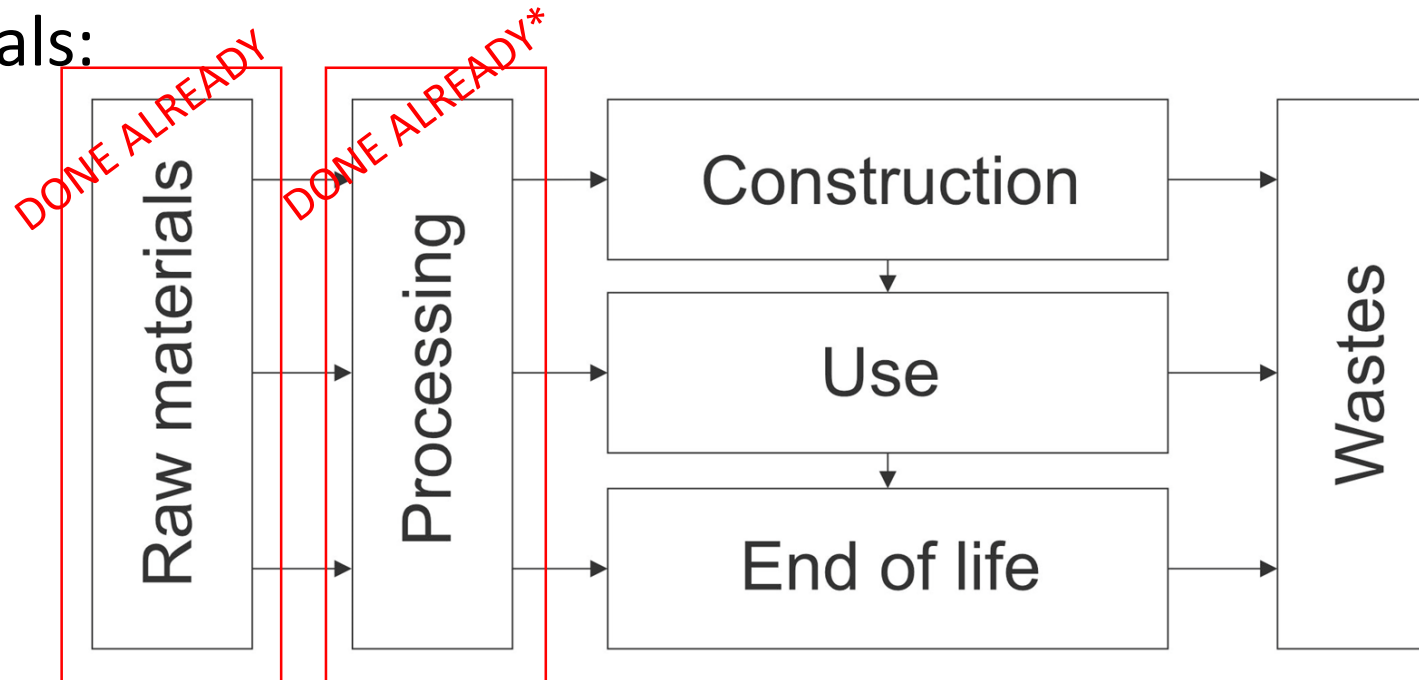
Wastes

- Wastes are anything from the construction, use and demolition of your product, process or infrastructure
- Wastes can include solid waste, atmospheric or water discharges, discharges to land or landfill
- Raw material extraction and processing will generate wastes as well
- Wastes considers what processes are undertaken to treat and discharge the wastes



Developing the flow chart

- In most cases, LCAs around raw material extraction and processing have already been done, so including details about these in the flowchart can be neglected
- Data for these can be found in various LCA databases (e.g. Gabi), and journals:

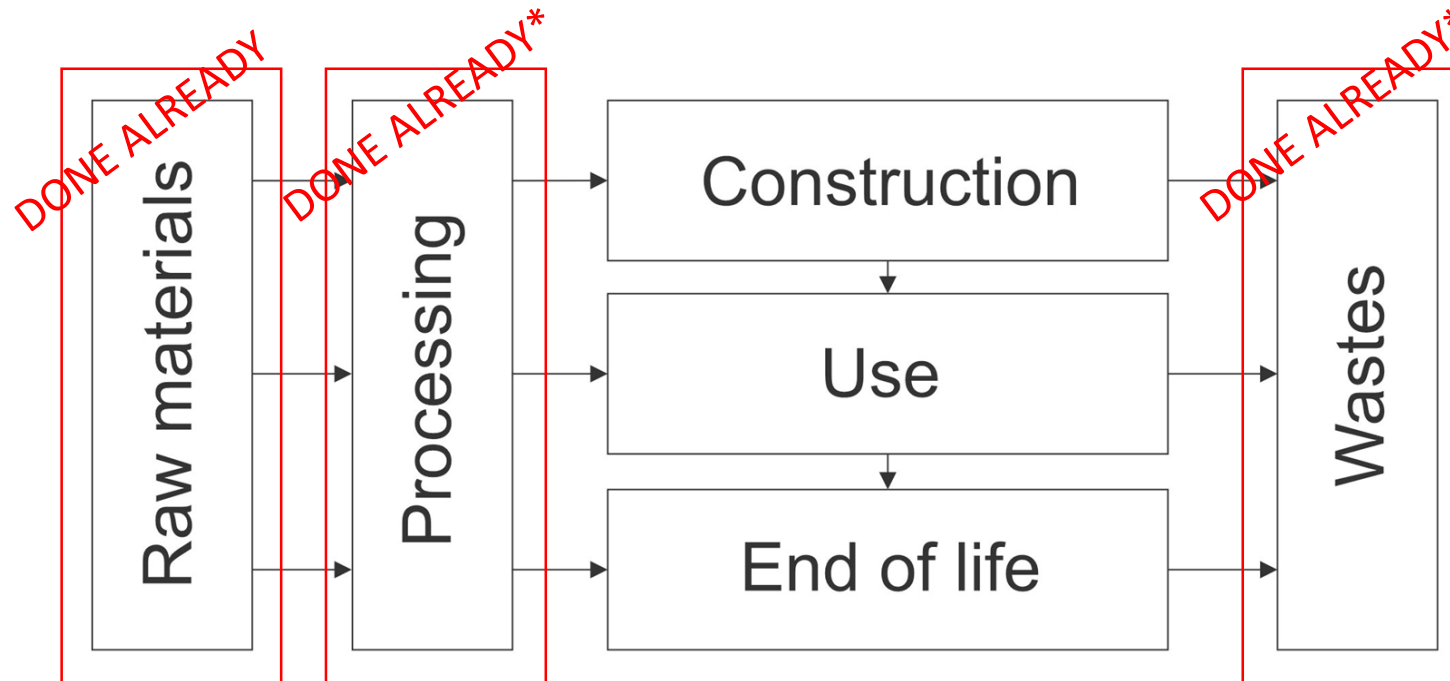


Developing the flow chart

- Should I focus on construction, use and or disposal?
 - This depends on where the focus of the LCA is and the bulk of the environmental impacts
 - E.g. an IC car – bulk of the impact is in the use of the car
 - E.g. an electric car – depends on the country and their electricity mix, the bulk of the impact might be in the fabrication of the car, but in places where electricity is generated by burning fossil fuels, some of the impact will be in the use of the car
 - E.g. a nuclear power plant – the bulk of the impact will be in the construction of the plant and extraction of uranium
 - Unless you are focusing specifically on different options for end of life, you can neglect end of life

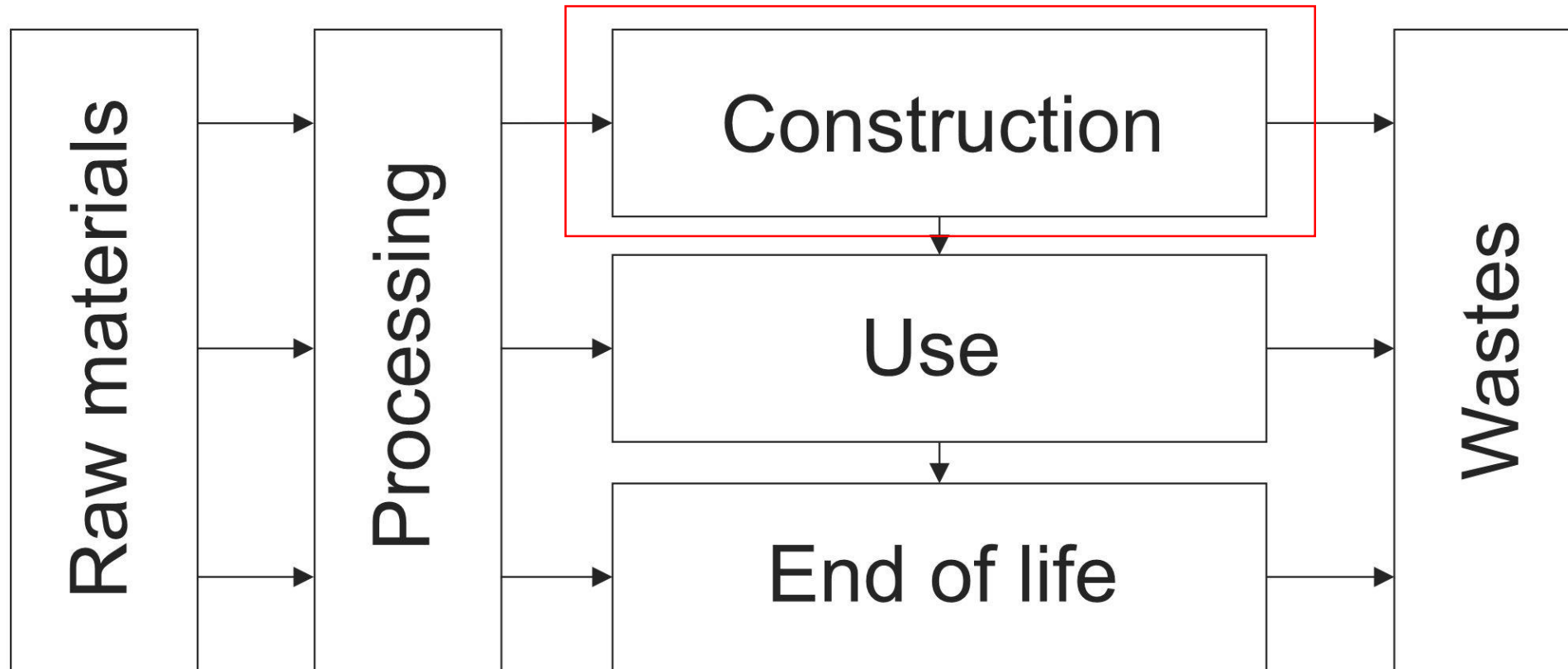
Developing the flow chart

- Should I consider wastes?
 - In many cases, impacts from processing/discharging wastes have already been done

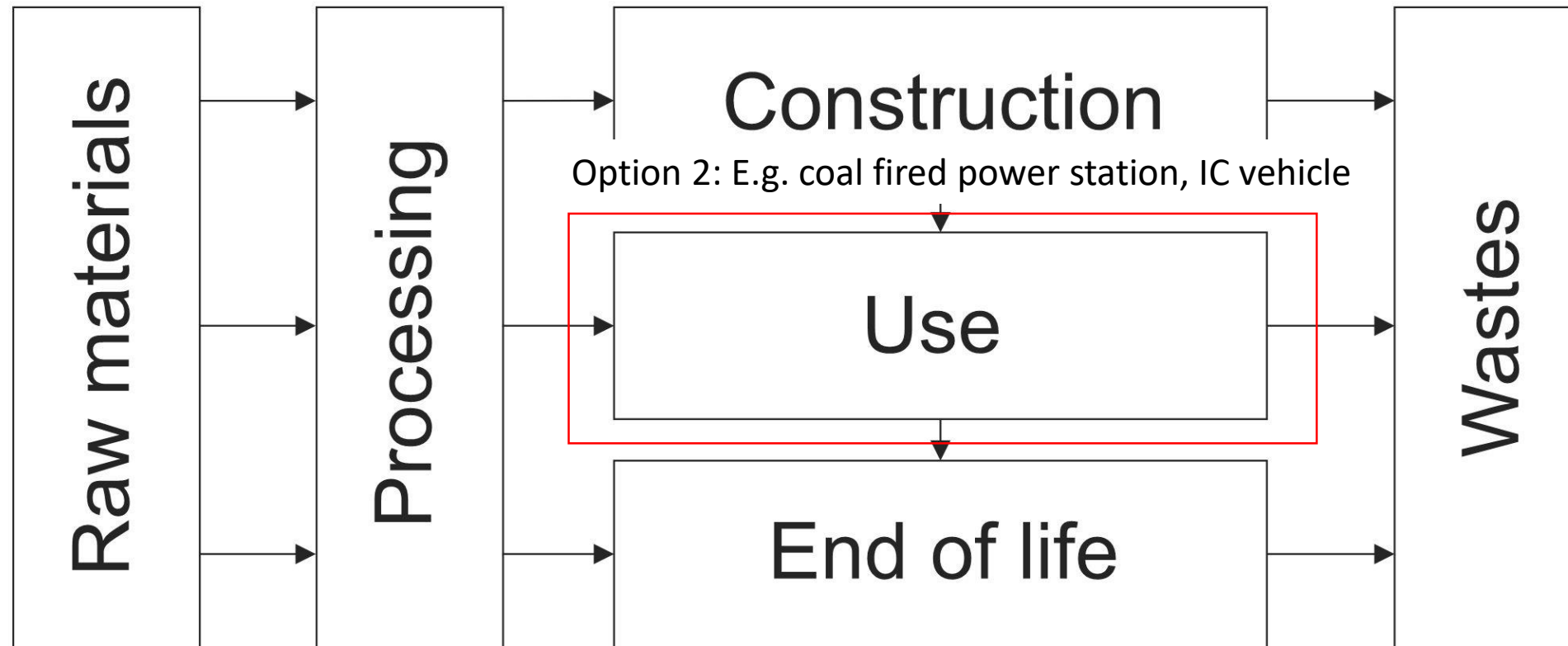


Where to focus efforts?

Option 1: E.g. hydro, solar, nuclear, bridges, may include some processing, will need to decide a life time

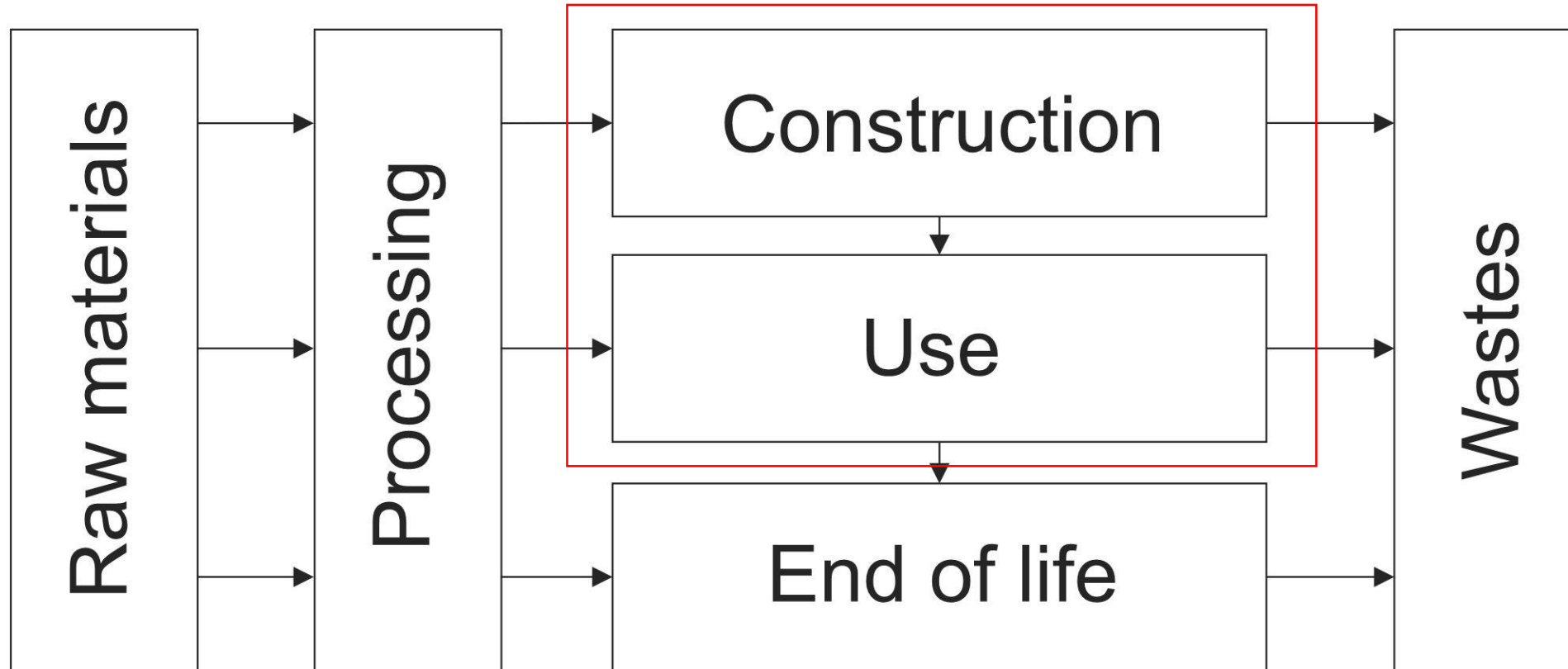


Where to focus efforts?

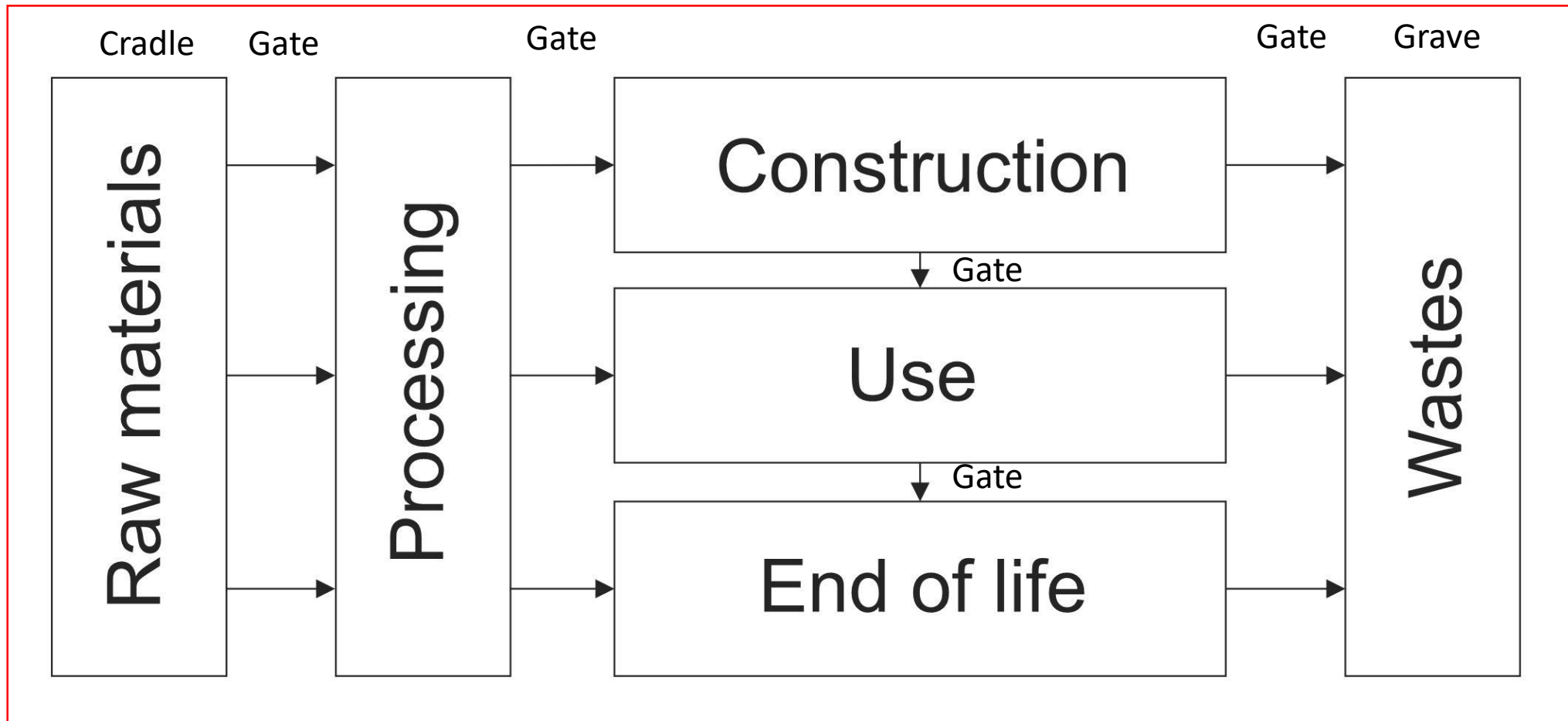


Where to focus efforts?

Option 3: E.g. Passive home heating, electric vehicle, may include some processing, will need to decide a life time

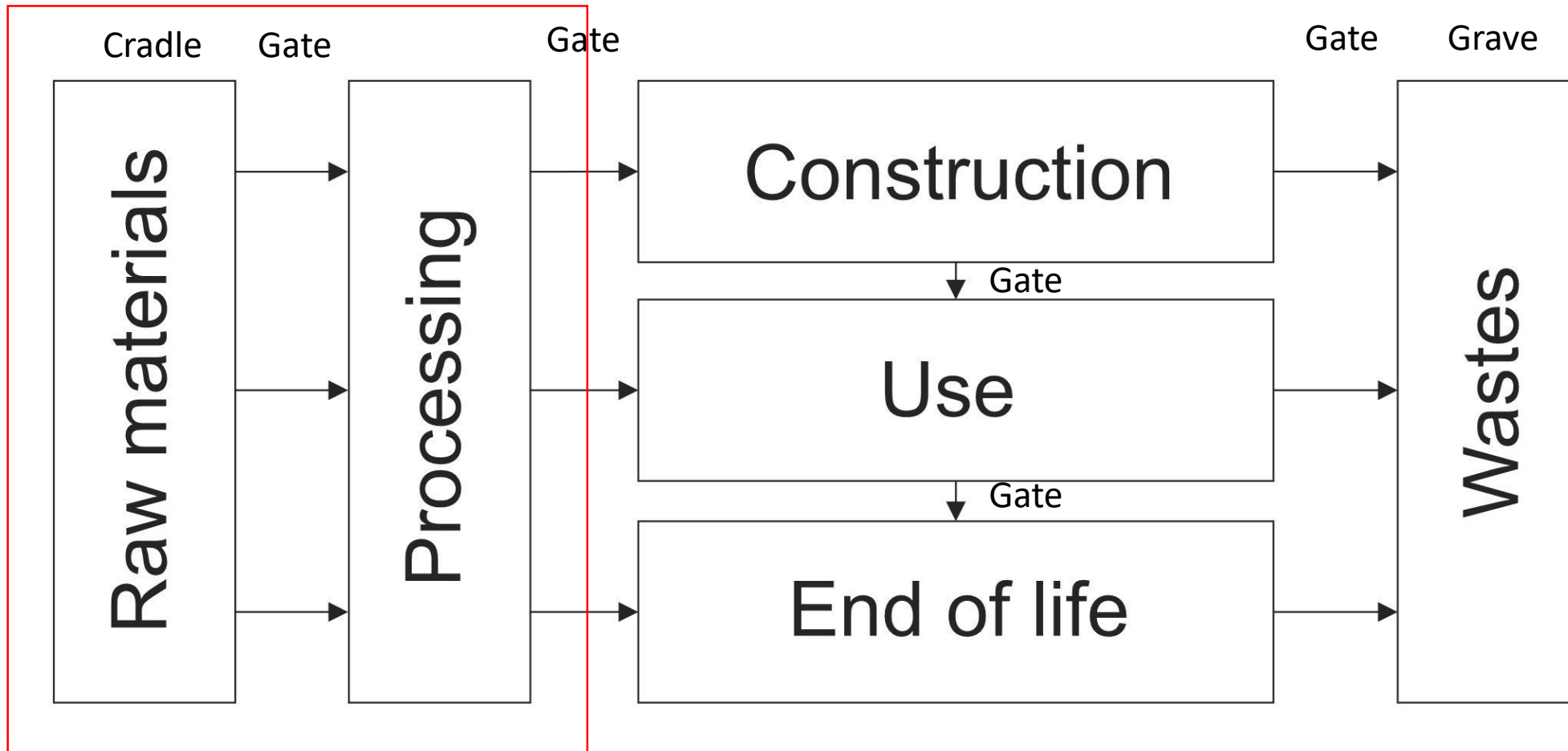


Types of LCA – Cradle to Grave



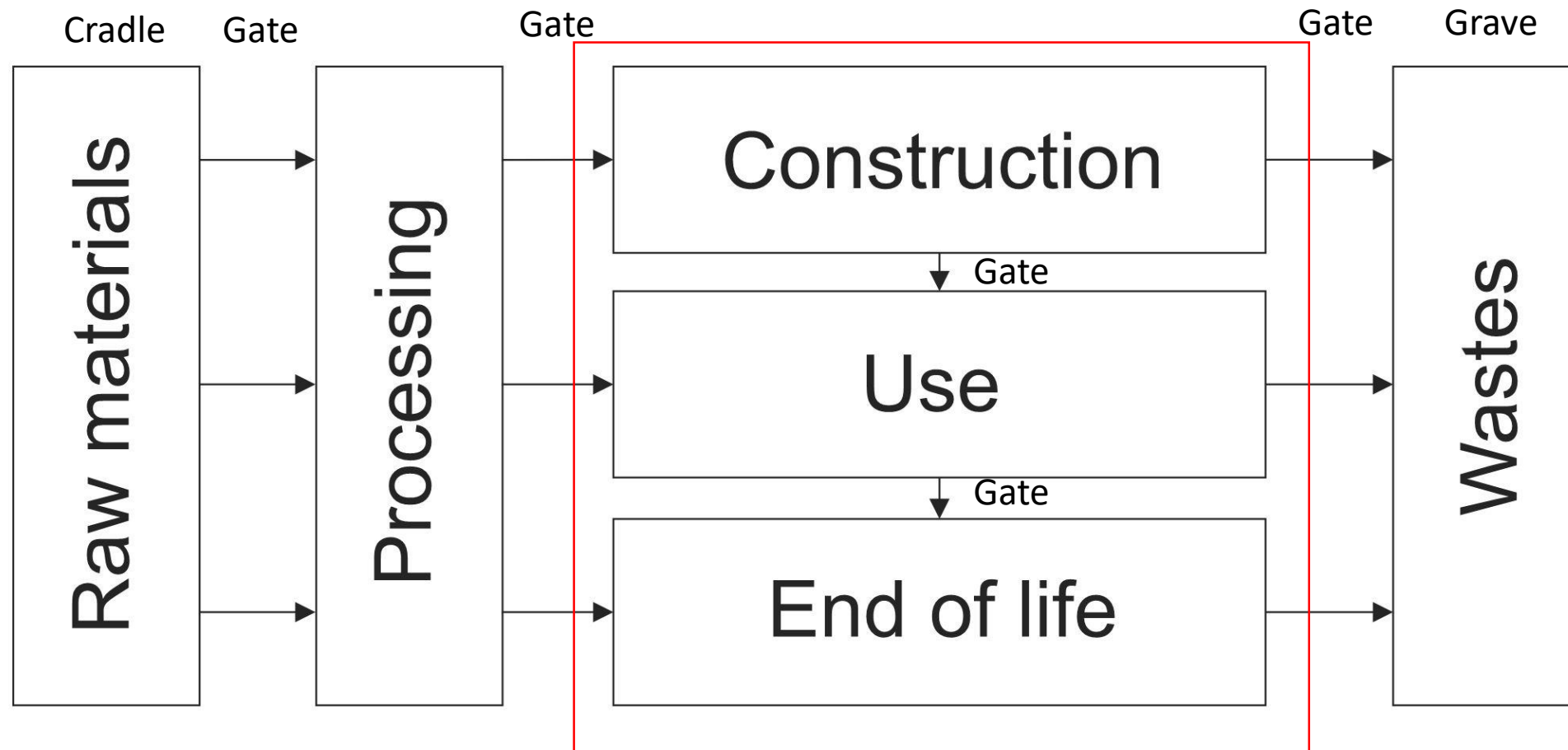
Types of LCA – Cradle to Gate

Can be up to any of the gates



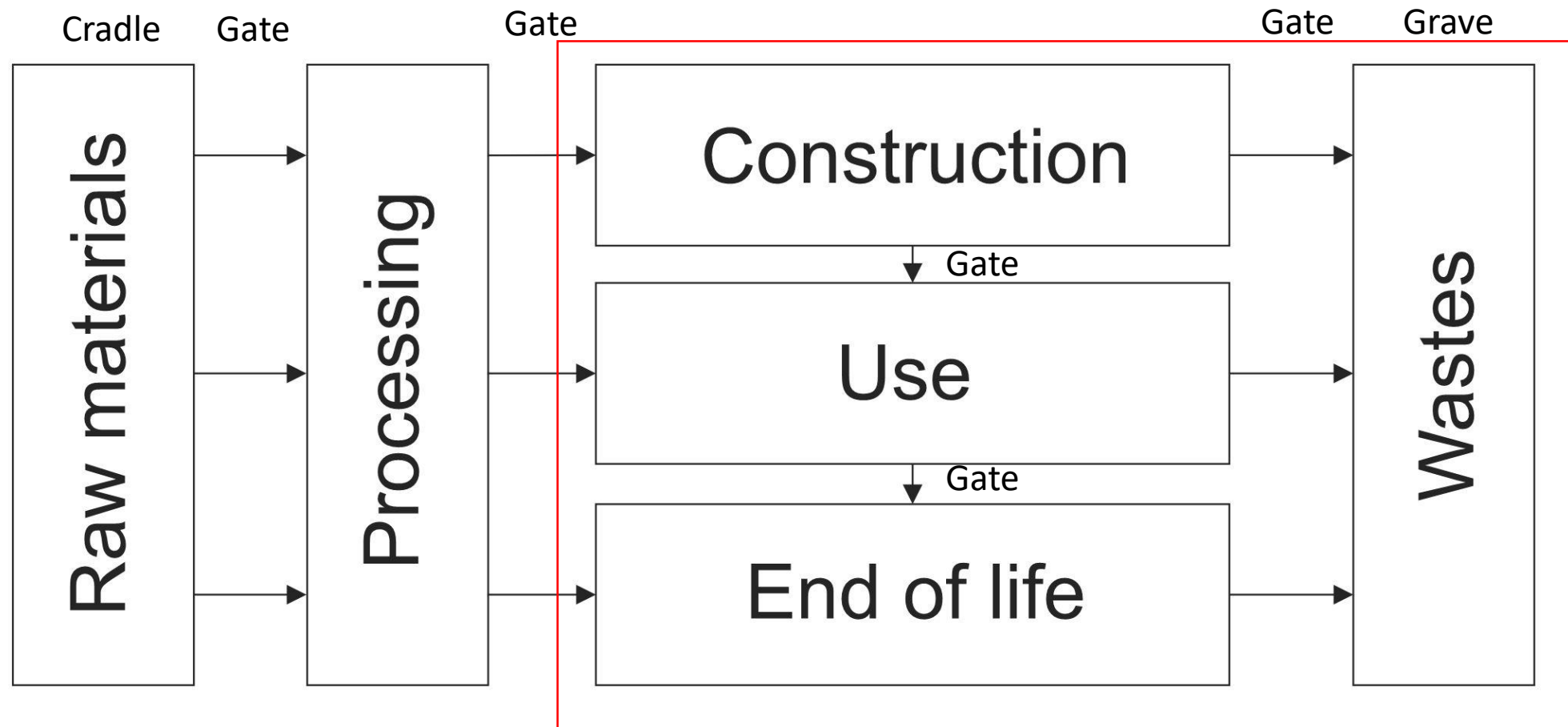
Types of LCA – Gate to Gate

Can be between any of the gates



Types of LCA – Gate to Grave

Can be from any of the gates



Developing a detailed flow chart

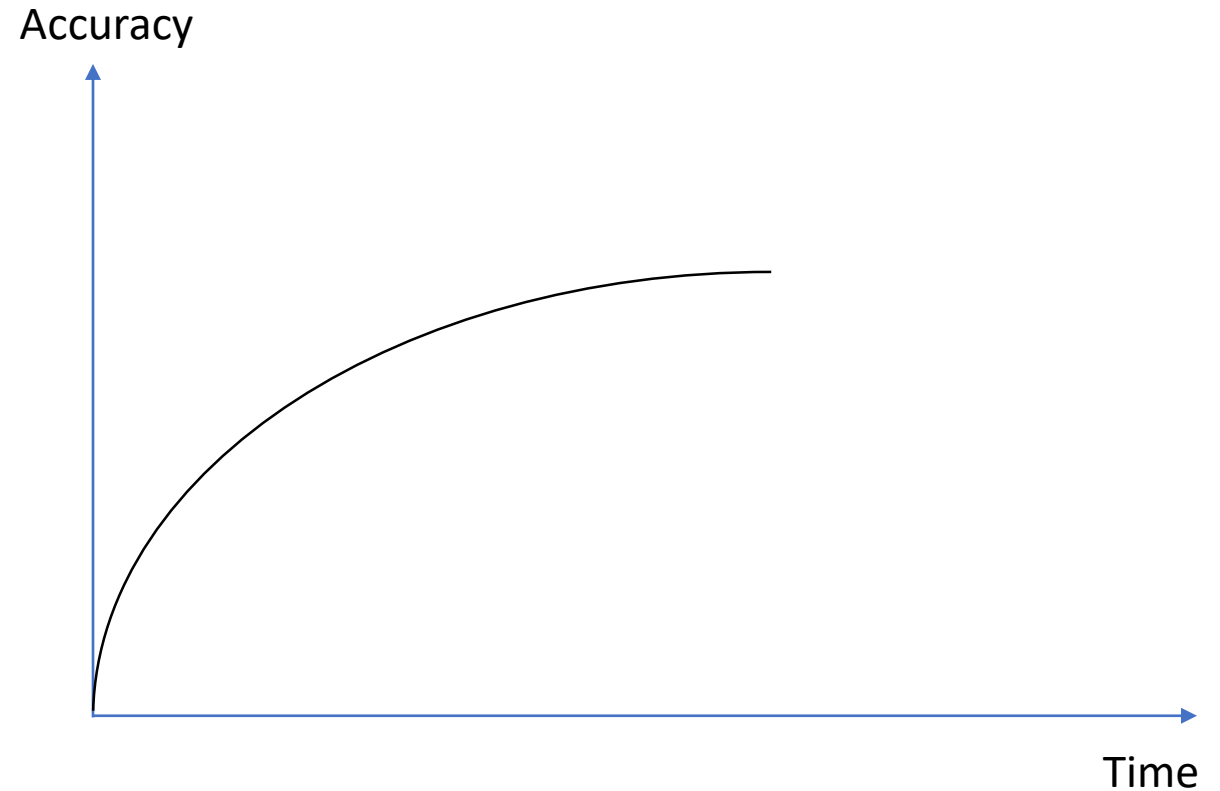
- Once the focus has been decided, you can provide more detail for the construction and/or use
- E.g. a watch:
 - Cutting billet (e.g. if aluminium) for the casing
 - Machining the billet to make the case (e.g. CNC machining)
 - Circuit board manufacture (might like to break this down into steps)
 - Production of motors, gears, dials, glass, buttons
 - Production of batteries
 - Assembly
 - Testing
 - Packaging

Developing a detailed flow chart

- E.g. a coal fired power plant
 - Transporting coal
 - Sizing coal
 - Pulverising coal
 - Blowing coal into the boiler
 - Blowing air into the boiler
 - Burning, ash collection
 - Heat recovery (super heaters, primary, reheater, economiser, air preheater)
 - Exhaust cleaning (electrostatic precipitator)
 - Maintenance (e.g. daily, weekly, monthly, yearly...)

Developing a detailed flow chart

- Where to get information?
 - Google
 - Wikipedia
 - Books (Library)
 - LCA reports
 - Manufacturers websites
 - Company data
 - Physically going there



Activity

- Draw a general flow diagram for your process, product, infrastructure
- Decide on the focus (scope) of your LCA
- Determine the construction and/or use steps (and any processing steps if needed) for your part of the LCA
- Expand the flow diagram to include those steps
- Consider what the material flows in and out might be for each step and add those in (you don't need quantities)
- What type of LCA is your part of your group LCA?